

1 Archetype 01_CH (0_1918)

Swiss CH building assumptions; reference weather ZUESTA, TIME_HORIZON=present, SCENARIO=none, occupancy SIA2024 (residential).

1.1 Geometry inputs

Component	Area [m²]	U-value [W/(m²·K)]	g-value [-]
Floor	95.9	0.9	—
Facade (opaque)	174	0.81	—
Roof	109	0.46	—
Windows	33.1	2	0.6

1.2 Thermal mass

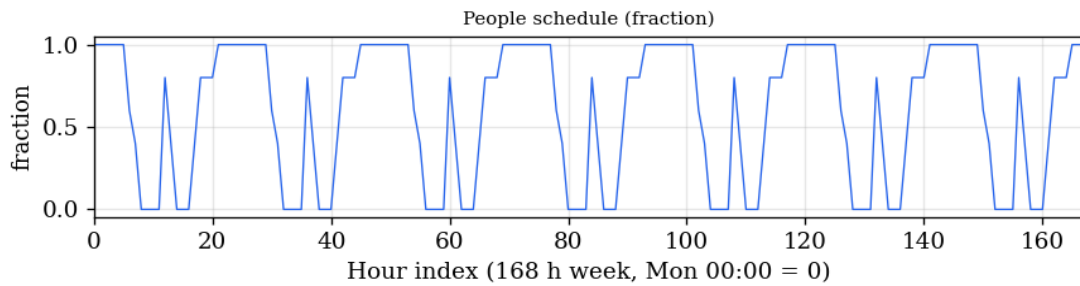
Quantity	Value
Heat capacity class (TABULA / ISO 13790)	heavy
Thermal mass capacitance [kWh/K]	16.11
Thermal mass per floor area [Wh/(m²·K)]	72.22
Conditioned floor area [m²]	223
TABULA building code	AT.N.SFH.01.Gen.ReEx.001.002
TABULA period label	0_1918

1.3 Internal loads

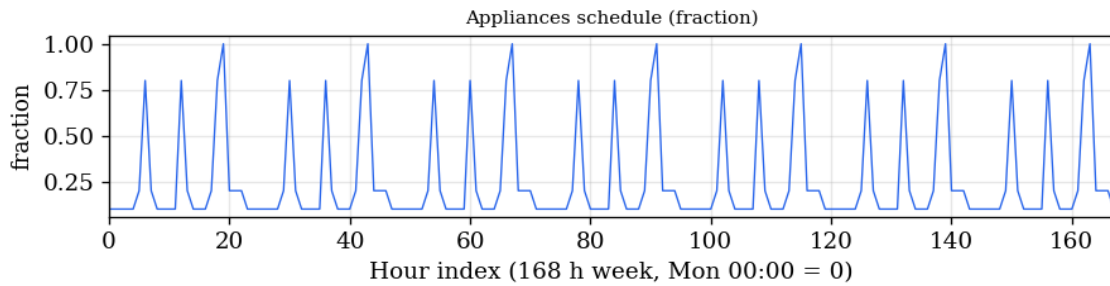
Load	Intensity	Schedule	Seasonal variation
People (sensible internal gains)	1.78 W/m² (0.02 pers/m² × 89.0 W/pers)	Weekly hourly profile (168 h, residential); sensible heat 89.0 W per person	Constant monthly multiplier on weekly profile: 0.8 (SIA presence utilisation)
Appliances (electricity → internal gains)	10 W/m²	Weekly hourly profile (168 h, residential)	Uniform annual multiplier on weekly profile: 0.7994610375028071 (SIA t_A,Ps vs daily equivalent)
Lighting (electricity → internal gains)	3 W/m²	SIA 2024 residential Nutzungsstunden (168 h weekly shape)	Annual full-load target t_L = 846.0 h; Monthly scale 1 + 0.35·cos(2π·m/12) (winter higher)

1.4 Weekly schedule profiles

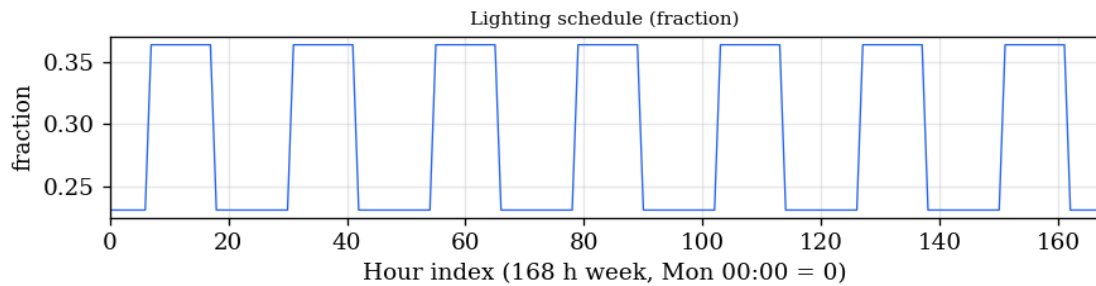
People schedule (weekly)



Appliances schedule (weekly)



Lighting schedule (weekly)



1.5 Ventilation

Mechanism	Parameter	Value / schedule
Infiltration (Swiss, replaces TABULA n _{air} infiltration)	Air change rate	0.3 1/h
Hygienic ventilation (SIA floor area)	Flow per conditioned area	0.6 m ³ /(h·m ²); enabled: yes
Occupancy-driven ventilation	Flow per person (when present)	29 m ³ /(h·person); enabled: yes; schedule follows SIA 2024 people profile
Summer window ventilation	Target ACH when active	2 1/h; enabled: yes
Summer window ventilation	Running-mean outdoor enable threshold [°C]	-10
Summer window ventilation	Open when outdoor cooler than operative	yes